

PRATHYUSHA T

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EDUCATION

S No.	Course	Institute	Board/University	Year of passing	Percentage
1	Ph.D (Chemical Engineering)	Indian Institute of Technology Tirupati	Indian Institute of Technology Tirupati	Ongoing	8.46
2	M.Tech (Environmental Engineering Offered by Chemical Engineering)	Jawaharlal Nehru Technological University Ananthapuram	Jawaharlal Nehru Technological University, Ananthapuram	2024	8.82
3	B. Tech (Agricultural Engineering)	Siddharth Institute of Engineering and Technology, Puttur	Jawaharlal Nehru Technological University, Ananthapuram	2021	89
4	Intermediate (MPC)	APSWR College, Rajapuram	Board of Intermediate Education AP	2017	83
5	SSC	APSWR School, Rajapuram	Board of Secondary Education AP	2015	83

JRF (Junior Research Fellow)

- Development of Semi-Automatic Farm Devices for usage in Small Agricultural Fields (April 2024 – December 2024)

INTERNSHIP

Medetronix Labs Pvt Ltd..

- Quality Analyst Intern & Content writer|Bangalore | Aug 2023 – Present

Content Writing:

- Created engaging web content for the company's website, effectively communicating key information to users.
- Developed user manuals and material safety data sheets providing comprehensive and accessible guidance during the internship.
- Generated detailed FAQ responses, enriching user comprehension and satisfaction.

Quality Testing and Evaluation:

- Conducted meticulous product testing procedures, including the evaluation and monitoring of electrode potential.

Research and Development (R&D) Support:

- Engaged in R&D activities, driving innovation and problem-solving initiatives to improve existing products and study the potential of new products.

PROJECT 1**Title: Development of Solar Power Operated Boom Sprayer**

This project aims to spray chemicals on plants using renewable energy sources. The study was undertaken to develop a prototype for the small-holding farmers with the purpose of reducing the drudgery and losses caused by using the conventional method of spraying. The study also included evaluating the prototype's operational parameters under local Agro-climatic conditions.

Roles and Responsibilities

- Taking the Lead and motivating the team.
- Prototype design using AutoCAD.
- Managing the operations.
- Solving problems.
- Taking care of the health and safety of team members.

PROJECT 2

At Advance Research Center International (ARCI) Hyderabad | March 2023 – August 2023

Title: Characterization and performance evaluation of EPDM rubber-based membranes for electrochemical water splitting and CO₂ reduction reactions

This project involves analyzing material properties, such as wettability, porosity, contact angle, and conductivity, using SEM and XRD to assess suitability for electrochemical water splitting and CO₂ reduction. Performance evaluation includes measuring efficiency, stability, and selectivity in these reactions, as well as gauging catalytic activity, gas permeability, and durability over extended cycles. Comprehensive testing determines the membrane's ability to facilitate efficient and selective electrochemical transformations for

sustainable water splitting and CO₂ reduction, which are pivotal for advancing clean energy technologies.

PATENT

- Pratyusha T, Dilip Kumar Behara, agricultural vehicle with Boom sprayer, Indian Patent Granted with Patent No: 433583-001 on 13 Jan 2026

PUBLICATIONS

Book Chapters

- T. Prathyusha, V. Sravanthi, Dilip Kumar Behara, Chapter 17 Biotechnologies for Environmental Sustainability: Advancing Clean Energy, Biodiversity, Water Security, and Climate Resilience, Edited by Ranbir Chander Sobti, Tejinder Kaur, Harpreet Walia, Puja Rattan, and Archana Narula, Academic Press, Elsevier, ISBN No: 978-0443-38325-0

Review Papers

- V Koti Bhanu Teja, Prathyusha T, Dilip Kumar Behara, Ultra-Low Emission Strategies in steel, Coking, and Cement Industries: Pathways to Decarbonization and Sustainability: Review Article Journal of Chemical Engineering (2025) Vol. 12
- T. Prathyusha, V Koti Bhanu Teja, K. Subba Rao, B. Dilip Kumar, Advancements in Membrane Technologies for Enhanced Water Splitting and CO₂ Reduction: A Comprehensive Review JCHR (2024) 14(2), 313-328 | ISSN:2251-6727

TECHNICAL SKILLS

- **Microsoft Tools:** Word, PowerPoint, Excel, and Outlook
- Research and Development
- Data Analysis
- Project management (Asana tool)
- Basics of Adobe Illustrator.

ACHIEVEMENTS

- Participated in a International Conference on “membrane processes for a green environment: current status, challenges and future prospects” (icmpge-2024) Jawaharlal Nehru Technological Institute, Anantapur(JNTUA), Andhra Pradesh

- Participated in a national conference on the millet revolution organized by the College of Agricultural Science and Applied Research (CASAR) Bharatiya Engineering Science and Technology Innovation University, Andhra Pradesh (invited speaker).
- Awarded Best Paper Presentation at "Sudhee 2023," a prestigious event held at Chaitanya Bharathi Institute of Technology, Hyderabad.
- Awarded Best Paper Presentation at "Pranathi 2023," an International Conference a prestigious event held at BVRIT, Narsapur.
- Presented an oral presentation at "ICRACACE-22", on 2nd international congress on recent advances in chemistry and chemical engineering held at Jawaharlal Nehru Technological University, Hyderabad
- Actively participated in the 10th National Level Technical Symposium at Siddharth Institute of Science and Technology, Puttur.
- Engaged in an enriching learning experience through participation in the International Webinar on "Emerging Technologies in Agricultural Engineering for Food Safety & Security," organized by Dr. NTR College of Agricultural Engineering, Bapatla.
- Exhibited scientific aptitude and passion for knowledge by participating in the renowned National Science Fair.

WORKSHOPS

- **Repair and overhauling of stationary I.C Engine** from Southern Region Farm Machinery Training and Testing Institute, Garladinne.
- **AutoCAD** by AP Skill Development Corporation.

DECLARATION

I, Prathyusha T, hereby declare that all the information furnished above is complete and true to the best of my knowledge and belief.

